A PROOFS

A.1 proof of Lemma. 4.3

Proof. Take any $t > 0$ and $x^* \in F_q$. Since $x^* \in \mathcal{M}_S$, we have $\mu(B(x^*, t)) > 0$, where $B(x^*, t)$ is an open ball in the ambient space.

Because μ is supported on $\mathcal{M}_S$,

$$\mu(B(x^*, t)) = \mu(B(x^*, t) \cap \mathcal{M}_S).$$

By the triangle inequality, $B(x^*, t) \cap \mathcal{M}_S \subseteq S_q(t)$. Hence, by monotonicity of μ,

$$0 < \mu(B(x^*, t) \cap \mathcal{M}_S) \le \mu(S_q(t)) = \mu_q(t).$$

Thus $\mu_q(t) > 0$, and $S_q(t)$ has positive probability.

Let$N_t = \sum_{i=1}^{n} \mathbf{1}\{X_i \in S_q(t)\}$. Since the samples are i.i.d., $\Pr(X_i \in S_q(t)) = \mu(S_q(t)) = \mu_q(t)$, so

$$N_t \sim \text{Binomial}(n, \mu_q(t)).$$

That is, N_t counts how many samples fall into the target region $S_q(t)$.

By the law of large numbers, we have $\frac{N_t}{n} \xrightarrow{\mathbb{P}} \mu_q(t) > 0$. Combining this with $k_n/n \to 0$, we obtain

$$\Pr(N_t \ge k_n) \to 1.$$

This event is equivalent to $r_{k_n,n}(q) \le \Delta_{\mathcal{M}_S}(q)+t$, and hence Lemma. 4.3 holds.

□

A.2 proof of Theorem. 4.4

Proof. Take any $\varepsilon > 0$ and define the bad set

$$K_\varepsilon = \{x \in \mathcal{M}_S : d_E(x, F_q) \ge \varepsilon\},$$

consisting of all points on the manifold whose distance to F_q is at least ε. If some point in the top-k_n set lies in K_ε, then the top-k_n set is not fully concentrated near F_q. If $K_\varepsilon = \emptyset$, then every point on the manifold is within distance ε of F_q, and the conclusion is immediate; hence it remains to consider the case $K_\varepsilon \neq \emptyset$.

We first note that K_ε is compact and disjoint from F_q. Indeed, $\mathcal{M}_S$ is compact, F_q is closed because it is the set of minimizers of the continuous function $\delta(x, q)$ on the compact set $\mathcal{M}_S$, and $d_E(x, F_q)$ is continuous, so K_ε is a closed subset of $\mathcal{M}_S$. Since a closed subset of a compact set is compact, K_ε is compact. Moreover, K_ε is disjoint from F_q by definition, since every point of F_q satisfies $d_E(x, F_q) = 0$.

Next, we show that the continuous function

$$f(x) = \delta(x, q) - \Delta(q)$$

attains a positive minimum on K_ε. For $x \in F_q$, we have $\delta(x, q) = \Delta(q)$, so $f(x) = 0$. For $x \in K_\varepsilon$, since $x \notin F_q$ and F_q is the set of points in $\mathcal{M}_S$ closest to q, we have $\delta(x, q) > \Delta(q)$, and hence $f(x) > 0$. Since K_ε is compact, f attains its minimum on K_ε; we denote this minimum by

$$\gamma_q(\varepsilon) = \min_{x \in K_\varepsilon} \big(\delta(x, q) - \Delta(q)\big).$$

Because $f(x) > 0$ for every $x \in K_\varepsilon$, we have $\gamma_q(\varepsilon) > 0$. Geometrically, any point at distance at least ε from F_q has distance to q at least $\gamma_q(\varepsilon)$ larger than the theoretical minimum distance $\Delta(q)$.

Now suppose that $N^*_{k_n}(q)$ contains some point $x \in K_\varepsilon$. Since x is one of the top-k_n points, its distance to q is at most the k_n-th nearest-neighbor distance. So $\delta(x, q) \le r_{k_n,n}(q)$. On the other hand, since $x \in K_\varepsilon$, we have $\delta(x, q) \ge \Delta(q) + \gamma_q(\varepsilon)$. Combining the two inequalities gives

$$r_{k_n,n}(q) \ge \delta(x, q) \ge \Delta(q) + \gamma_q(\varepsilon),$$

and therefore $r_{k_n,n}(q) - \Delta(q) \ge \gamma_q(\varepsilon)$. Thus, if any point in the top-k_n set is at distance at least ε from F_q, then the gap between the k_n-th nearest-neighbor distance and the theoretical minimum distance is at least $\gamma_q(\varepsilon)$.

Finally, we complete the proof using Lemma. 4.3, which gives

$$r_{k_n,n}(q) - \Delta(q) \xrightarrow{\mathbb{P}} 0.$$

In particular, for the positive number $\gamma_q(\varepsilon)/2$,

$$\Pr\left(r_{k_n,n}(q) - \Delta(q) > \frac{\gamma_q(\varepsilon)}{2}\right) \to 0.$$

Consider the event

$$\left\{\max_{x \in N^*_{k_n}(q)} d_E(x, F_q) \ge \varepsilon\right\}.$$

If this event occurs, then at least one point $x \in N^*_{k_n}(q)$ satisfies $d_E(x, F_q) \ge \varepsilon$, i.e., $x \in K_\varepsilon$. By the previous step, this implies

$$r_{k_n,n}(q) - \Delta(q) \ge \gamma_q(\varepsilon) > \frac{\gamma_q(\varepsilon)}{2},$$

so

$$\left\{\max_{x \in N^*_{k_n}(q)} d_E(x, F_q) \ge \varepsilon\right\} \subseteq \left\{r_{k_n,n}(q) - \Delta(q) > \frac{\gamma_q(\varepsilon)}{2}\right\}.$$

Therefore,

$$\Pr\left[\max_{x \in N^*_{k_n}(q)} d_E(x, F_q) \ge \varepsilon\right] \le \Pr\left[r_{k_n,n}(q) - \Delta(q) > \frac{\gamma_q(\varepsilon)}{2}\right] \to 0.$$

Since this holds for every $\varepsilon > 0$, Theorem 4.4 is proved □

A.3 proof of Lemma. 4.5

Proof. Since $q_{\text{id}} \in \mathcal{M}_S$, we have $\Delta(q_{\text{id}}) = 0$, and the distance margin is $h_{q_{\text{id}}}(x) = \delta(x, q_{\text{id}})$. Thus the sublevel set is a local ball on the manifold:

$$S_{q_{\text{id}}}(r) = \mathcal{M}_S \cap B(q_{\text{id}}, r).$$

Near q_{id}, the manifold locally resembles an m-dimensional Euclidean space. Hence the volume of this local ball grows as the m-th power of the radius:

$$\text{Vol}\big(\mathcal{M}_S \cap B(q_{\text{id}}, r)\big) \asymp r^m.$$

Since the density is uniformly η, the probability mass also grows as r^m:

$$\mu_{q_{\text{id}}}(r) = \eta \times \text{Vol}\big(\mathcal{M}_S \cap B(q_{\text{id}}, r)\big) \asymp r^m.$$

Now with n i.i.d. samples, the number of samples falling into this ball is approximately

$$n\mu_{q_{\text{id}}}(r) \asymp nr^m.$$

Since the expected number of samples falling into the ball equals k_n, we have $nr^m_{\text{id}} \approx k_n$. Therefore,

$$r_{\text{id}} \asymp \left(\frac{k_n}{n}\right)^{1/m}.$$

To make this rigorous, note that for any $r > 0$,

$$N_r = \sum_{i=1}^{n} \mathbf{1}\{X_i \in \mathcal{M}_S \cap B(q_{\mathrm{id}}, r)\} \sim \mathrm{Binomial}(n, \mu_{q_{\mathrm{id}}}(r)).$$

The event $r_{k_n,n}(q_{\mathrm{id}}) > r$ is equivalent to $N_r < k_n$, so

$$\Pr\big(r_{k_n,n}(q_{\mathrm{id}}) > r\big) = \Pr\big(N_r < k_n\big).$$

From the volume and density estimates, there exist constants $c_1, c_2 > 0$ such that for sufficiently small r,

$$c_1 r^m \le \mu_{q_{\mathrm{id}}}(r) \le c_2 r^m.$$

For the upper bound, take $r_+ = C(k_n/n)^{1/m}$ with $C > 0$ sufficiently large so that

$$n\mu_{q_{\mathrm{id}}}(r_+) \ge c_1 C^m k_n \ge (1+\alpha)k_n$$

for some fixed $\alpha > 0$. By the Chernoff lower-tail bound,

$$\Pr\big(N_{r_+} < k_n\big) \le \exp\left(-\frac{\alpha^2}{2(1+\alpha)} k_n\right) \to 0.$$

Hence

$$\Pr\left(r_{k_n,n}(q_{\mathrm{id}}) \le C\left(\frac{k_n}{n}\right)^{1/m}\right) \to 1.$$

For the lower bound, take $r_- = c(k_n/n)^{1/m}$ with $c > 0$ sufficiently small so that

$$n\mu_{q_{\mathrm{id}}}(r_-) \le c_2 c^m k_n \le (1-\alpha)k_n$$

for some fixed $\alpha \in (0,1)$. By the Chernoff upper-tail bound,

$$\Pr\big(N_{r_-} \ge k_n\big) \le \exp\left(-\frac{\alpha^2}{2} n\mu_{q_{\mathrm{id}}}(r_-)\right) \to 0.$$

Hence

$$\Pr\left(r_{k_n,n}(q_{\mathrm{id}}) \ge c\left(\frac{k_n}{n}\right)^{1/m}\right) \to 1.$$

Combining the two bounds, there exist constants $c, C > 0$ such that

$$\Pr\left(c\left(\frac{k_n}{n}\right)^{1/m} \le r_{k_n,n}(q_{\mathrm{id}}) \le C\left(\frac{k_n}{n}\right)^{1/m}\right) \to 1,$$

which is exactly $r_{k_n,n}(q_{\mathrm{id}}) = \Theta_P\left(\left(\frac{k_n}{n}\right)^{1/m}\right)$ □

A.4 proof of Lemma. 4.6

Proof. Take a local coordinate chart near x^*:

$$\phi : U \subseteq \mathbb{R}^{m-1} \times \mathbb{R}_+ \to \mathcal{M}_S, \quad \phi(0,0) = x^*.$$

Write the local coordinates as $(z,s) \in U$, where $z \in \mathbb{R}^{m-1}$ is along the intrinsic boundary direction of the manifold, $s \ge 0$ denotes the depth into the interior of the manifold, and $s = 0$ corresponds to the boundary. By the nondegeneracy assumption, there exist constants $c, C > 0$ such that

$$c\big(s + \|z\|^2\big) \le h_q\big(\phi(z,s)\big) \le C\big(s + \|z\|^2\big) \tag{3}$$

holds in a sufficiently small coordinate neighborhood.

Hence, $h_q(\phi(z,s)) \asymp s + \|z\|^2$. The sublevel set is

$$S_q(t) = \{x \in \mathcal{M}_S : h_q(x) \le t\}.$$

In local coordinates,

$$\phi(z,s) \in S_q(t) \iff h_q(\phi(z,s)) \le t.$$

From the lower bound in Eq. (3),

$$c(s + \|z\|^2) \le h_q(\phi(z,s)) \le t,$$

so $s + \|z\|^2 \lesssim t$; the reverse direction follows from the upper bound in (8). Hence, in local coordinates, the sublevel set is approximately

$$S_q(t) \approx \left\{\phi(z,s) : s \ge 0, \quad \|z\| \lesssim \sqrt{t}, \quad 0 \le s \lesssim t - \|z\|^2\right\}.$$

This is a thin cap: its width along the boundary direction z is about $\sqrt{t}$, while its depth into the manifold interior s is about t. Since $\sqrt{t} \gg t$ as $t \to 0$, the thin cap spreads widely along the boundary but is very thin toward the interior.

In local coordinates (z,s), the volume element is $dz\,ds$. The m-dimensional volume of the thin cap is

$$H^m(S_q(t)) \asymp \int_{\|z\| \lesssim \sqrt{t}} \left(\int_0^{t-\|z\|^2} ds\right) dz.$$

The inner integral equals

$$\int_0^{t-\|z\|^2} ds = t - \|z\|^2.$$

Let $z = \sqrt{t}\,u$. Then $dz = t^{(m-1)/2}\,du$ and $\|z\|^2 = t\|u\|^2$. Substituting,

$$\int_{\|z\| \lesssim \sqrt{t}} \big(t - \|z\|^2\big)\,dz = t \cdot t^{(m-1)/2} \int_{\|u\| \lesssim 1} \big(1 - \|u\|^2\big)\,du.$$

The remaining integral $\int_{\|u\| \lesssim 1} \big(1 - \|u\|^2\big)\,du$ is a positive constant independent of t. Hence

$$H^m(S_q(t)) = \Theta\left(t^{(m+1)/2}\right).$$

Since the density is uniformly η, the probability mass satisfies

$$\mu_q(t) = \eta \times H^m(S_q(t)) = \Theta\left(t^{(m+1)/2}\right).$$

As in the ID query derivation, equating the expected number of samples in this region to k_n and applying Chernoff bounds yields

$$T_{k_n,n}(q) = \Theta_P\left(\left(\frac{k_n}{n}\right)^{2/(m+1)}\right).$$

It remains to determine the tangential width and depth scale. Since $T_{k_n,n}(q) \to 0$, the sublevel set $S_q(T_{k_n,n}(q))$ lies entirely inside the local coordinate chart $\phi(U)$ with high probability. On this event, define

$$r_{\mathrm{tan}}^{\mathrm{cap}} = \sup\left\{\|z\| : \phi(z,s) \in S_q(T_{k_n,n}(q))\right\},$$
$$r_{\mathrm{depth}}^{\mathrm{cap}} = \sup\left\{s : \phi(z,s) \in S_q(T_{k_n,n}(q))\right\}.$$

On the complementary event, we may set both quantities to 1; since the complementary event has probability tending to zero, this does not affect the probabilistic order.

From the local relation $s + \|z\|^2 \lesssim T_{k_n,n}(q)$, setting $s = 0$ gives

$$\|z\|^2 \lesssim T_{k_n,n}(q) \quad \Longrightarrow \quad \|z\| \lesssim \sqrt{T_{k_n,n}(q)}.$$

Therefore,

$$r_{\mathrm{tan}}^{\mathrm{cap}} = \Theta_P\left(\sqrt{T_{k_n,n}(q)}\right) = \Theta_P\left(\left(\frac{k_n}{n}\right)^{1/(m+1)}\right).$$

Similarly, setting $z = 0$ gives

$$s \lesssim T_{k_n,n}(q),$$

and hence

$$r_{\text{depth}}^{\text{cap}} = \Theta_P\left(T_{k_n,n}(q)\right) = \Theta_P\left(\left(\frac{k_n}{n}\right)^{2/(m+1)}\right).$$

This proves Lemma. 4.6.

□

A.5 proof for Theorem. 4.7

Proof. Let D_{id} and D_{ood} denote the pairwise distances of two uniformly sampled points in an ID target region and an OOD target region, respectively. For k_n sampled points, the expected average pairwise distance equals the expected pairwise distance of two such points, so it suffices to compare $\mathbb{E}D_{\text{id}}$ and $\mathbb{E}D_{\text{ood}}$.

By Lemma 4.5, the ID target region is locally isotropic with characteristic radius $r_{\text{id}} = \Theta_P\left(\left(\frac{k_n}{n}\right)^{1/m}\right)$.Since all local directions scale at the same order, $\mathbb{E}D_{\text{id}} = \Theta_P(r_{\text{id}})$. By Lemma 4.6, the OOD target region is a thin cap with

$$r_{\tan}^{\text{cap}} = \Theta_P\left(\left(\frac{k_n}{n}\right)^{1/(m+1)}\right), \quad r_{\text{depth}}^{\text{cap}} = \Theta_P\left(\left(\frac{k_n}{n}\right)^{2/(m+1)}\right)$$

.

Now consider two uniformly sampled points P, Q in the OOD cap. In local coordinates, decompose their displacement into tangential and normal components: $P - Q = \Delta_{\parallel} + \Delta_{\perp}$. Then

$$D_{\text{ood}} = \|P - Q\| = \sqrt{\|\Delta_{\parallel}\|^2 + \|\Delta_{\perp}\|^2}.$$

Under the nondegenerate boundary-projection assumption, the tangential separation has typical scale $\Theta_P(r_{\tan}^{\text{cap}})$, while the normal separation is only $O_P(r_{\text{depth}}^{\text{cap}})$. Hence

$$\Theta_P(r_{\tan}^{\text{cap}}) = \mathbb{E}\|\Delta_{\parallel}\| \le \mathbb{E}D_{\text{ood}},$$

$$\mathbb{E}D_{\text{ood}} \le \mathbb{E}\|\Delta_{\parallel}\| + \mathbb{E}\|\Delta_{\perp}\| = \Theta_P(r_{\tan}^{\text{cap}}) + O_P(r_{\text{depth}}^{\text{cap}}) = \Theta_P(r_{\tan}^{\text{cap}}).$$

because $r_{\text{depth}}^{\text{cap}} = o(r_{\tan}^{\text{cap}})$. Therefore

$$\mathbb{E}D_{\text{ood}} = \Theta_P(r_{\tan}^{\text{cap}}) = \Theta_P\left(\left(\frac{k_n}{n}\right)^{1/(m+1)}\right).$$

Consequently,

$$\frac{\mathbb{E}D_{\text{ood}}}{\mathbb{E}D_{\text{id}}} = \Theta_P\left(\frac{r_{\tan}^{\text{cap}}}{r_{\text{id}}}\right) = \Theta_P\left(\left(\frac{k_n}{n}\right)^{-1/[m(m+1)]}\right) \longrightarrow \infty,$$

since $k_n/n \to 0$. Hence

$$\mathbb{E}D_{\text{ood}} \gg \mathbb{E}D_{\text{id}}.$$

The same asymptotic comparison holds for the expected average pairwise distance among k_n sampled points. Therefore the expected pairwise distance among OOD neighbors is asymptotically larger than that among ID neighbors. □

A.6 proof for Lemma. 4.9

Proof. We first characterize the survival probability of a candidate edge (u, v). Let $r = \delta(u, v)$, and define the lune

$$L(u, v) = B(u, r) \cap B(v, r).$$

By the MRNG pruning rule (Definition 4.8), the edge (u, v) survives if and only if no previously admitted neighbor of u lies in $L(u, v)$. Indeed, every point $w \in L(u, v)$ satisfies $\delta(w, u) < r$, and is therefore examined before v in the ascending-distance order. If such a point has already been admitted, it prunes (u, v).

Let $V_L(r)$ denote the volume of this lune. Since it is the intersection of two radius-r balls whose centers are distance r apart, scaling yields

$$V_L(r) = c_m r^m,$$

where $c_m > 0$ depends only on the ambient dimension m. Under local uniform sampling, the expected number of other sample points in the lune is thus

$$\mathbb{E}[X(u, v)] = \Theta(nr^m).$$

Under the assumed local random sampling model, the number of admitted points in the lune has the same order of intensity. In particular, the corresponding void probability satisfies

$$\Pr(Y(u, v) = 0) = \exp(-\Theta(nr^m)),$$

where $Y(u, v)$ counts the previously admitted neighbors of u that lie in $L(u, v)$. Therefore,

$$\Pr((u, v) \in E_{\text{MRNG}}) = \Pr(Y(u, v) = 0) = \exp(-\Theta(nr^m)).$$

Summing the survival probabilities over all ordered pairs of distinct points in $\mathcal{N}_k(q)$ and applying linearity of expectation gives Eq.(2).

We next compare ID and OOD target areas. By the geometric analysis in Section 4.1, the characteristic pairwise distance in an ID target area is of order r_{id}, whereas that in an OOD target area is governed by the tangential scale $r_{\tan}^{\text{cap}}$. Since $r_{\text{id}} \ll r_{\tan}^{\text{cap}}$, the corresponding pruning exponents satisfy

$$n\left(r_{\tan}^{\text{cap}}\right)^m \gg nr_{\text{id}}^m.$$

Hence, for comparable numbers of candidate ordered pairs and under a sufficiently uniform separation of the pairwise-distance scales, the survival probabilities of OOD candidate edges are exponentially smaller than those of ID candidate edges. Summing over the candidate pairs in Eq.(2) establishes the whole Therome 4.9. □